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Universal statistics of hippocampal place fields across species and dimensionalities

Highlights

- We present a model that explains irregular CA1 responses in large environments
- The theory explains statistics of field sizes, arrangement, and shape heterogeneities
- Predictions are quantitative, in 1/2/3D spaces of varying size, in rodents and bats
- CA1 inputs are modeled as Gaussian processes, indicative of random connectivity

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In brief

Mainali et al. show that a simple mathematical model explains recent observations of heterogeneous and irregular spatial responses in the hippocampus in large environments. The model unifies observations from environments of varying sizes and dimensions in multiple species, suggesting that a common mechanism underlies the hippocampal spatial code.



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Universal statistics of hippocampal place fields across species and dimensionalities

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SUMMARY

Hippocampal place cells have single, bell-shaped place fields in small environments. Recent experiments, however, reveal that, in large environments, place cells have multiple fields with heterogeneous shapes and sizes. We show that this diversity is explained by a surprisingly simple mathematical model, in which place fields are generated by thresholding a realization of a random Gaussian process. The model captures the statistics of field arrangements and generates new quantitative predictions about the statistics of field shapes and topologies. These predictions are quantitatively verified in bats and rodents, in one, two, and three dimensions, in both small and large environments. These results imply that common mechanisms underlie the diverse statistics observed in different experiments and further suggest that synaptic projections to CA1 are predominantly random.

INTRODUCTION

A central goal of systems neuroscience is to characterize how neural systems represent information about the world or about the brain's internal state. For several decades, much of the thinking about neural population codes was motivated by reports on neurons with highly stereotyped tuning functions. Neurons were often observed to have a smooth, typically unimodal tuning to the encoded variable, centered on preferred stimuli that vary across the neural population.[1,2] Examples include the radially symmetric receptive fields of retinal ganglion cells,[3] the receptive fields of simple cells in the primary visual cortex,[4,5] cosine tuning to movement direction in the motor cortex,[6] the unimodal fields of head direction cells,[7] and the fields of hippocampal place cells in small environments.[8]

Experiments in the past decade, however, have uncovered neural response functions that are much less stereotyped and regular than observed previously.[9–17] These recent findings were enabled by high-throughput recording techniques, which substantially reduced bias in the selection of cells for analysis, as well as new technologies that enabled the monitoring of neural activity in freely behaving animals, under reduced behavioral constraints.[18–21]

Some of the most striking examples of irregular neural responses were recently identified in area CA1 of the hippocampus. The classical view of spatial coding by place cells in this area has been that they are active in a single, compact region of space and exhibit a stereotyped bell-shaped tuning to position.[8,22] Several recent experiments in bats and rodents have revealed, however, that this picture breaks down in large environments.[9,10,21,23–25] Place cells typically fire in multiple locations, and furthermore, the multiple firing fields of individual cells, as well as those of the whole population, vary in size and shape, the latter of which can deviate substantially from the classical bell-shaped form. It remains unknown, however, whether the irregular statistics of place fields in bats and rodents can be described in terms of common mathematical principles and whether these statistics yield insights on the synaptic architecture that underlies spatial coding in hippocampal area CA1.

A number of theoretical studies have modeled the spatial selectivity of place cells in large environments. In one approach, heterogeneous activity patterns were embedded in the attractor manifolds of recurrent neural networks, serving as models of place-cell dynamics in area CA3. Place cells were modeled as having multiple, randomly positioned firing fields[26,27] with variable peak firing rates and field sizes.[27] Another study[28] predicted an exponential distribution of place-field sizes based on a normative theory. Finally, multi-peaked, multi-scale tuning curves were shown to be advantageous for coding efficiency.[21,29] However, there is a lack of a theory that can explain

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the observed heterogeneity of place-field properties, across the diverse experimental conditions, from basic principles.

Here, we report that a surprisingly simple generative model accounts for highly detailed features of place-field statistics. We model place fields as derived from a realization of a random Gaussian process over the spatial coordinates: a realization of the process is sampled for each cell and is then thresholded and rectified, resulting in multiple, heterogeneous fields (Figure 1A). A Gaussian process is a random function whose values at any discrete set of positions are jointly normal.[30] The statistics of a Gaussian process are uniquely determined by its mean, which we set to zero without loss of generality as its choice is redundant with the choice of the threshold, and by the spatial correlation function, which we assume to be translationally invariant.

Several lines of reasoning have led us to examine the above model. First, a Gaussian process is the random process that maximizes entropy under constraints on the mean and the spatial covariance function. Hence, our modeling choice invokes minimal assumptions other than a specification of a place cell's underlying input correlation structure.[31] Second, Gaussian statistics arise when many independent random variables are summed. The summed presynaptic input into area CA1 from area CA3 and entorhinal cortex is thus expected to be approximately Gaussian if synaptic weights are predominantly random (Figure 1; Methods S1 and S2). Third, a neural code structured as a random Gaussian process can achieve exponential coding efficiency.[29] We further discuss these motivations in the discussion.

RESULTS

Field arrangements

Place-field locations in the model are identified as the regions in space in which the Gaussian process exceeds a given threshold (Methods S1). Extensive results exist for the statistics of these regions in the mathematical literature, where they are called "excursion sets."[30] A recent result of central importance to this work is that threshold-crossing statistics assume a universal form when the spatial correlation function of the process obeys mild smoothness requirements and the threshold is sufficiently high.[32] Threshold-crossing statistics are then insensitive to the detailed structure of the correlation function and depend only on two scalar parameters: the normalized threshold, θ, defined as the ratio between the threshold and the standard deviation of the Gaussian process (Equation 3) and a correlation length, σ, which is derived from the spatial correlation function (Equation 4).

Specifically, the mean density of fields and the mean field size can both be expressed in terms of σ and θ using the Kac-Rice formula[33] for the threshold crossings of a Gaussian process (Methods S3, see Equations S12 and S14). Furthermore, the full distribution of field sizes acquires a universal form in the limit of large θ (Methods S4), expressed as

$$P(s) = \frac{2\beta}{D} s^{\left(\frac{2}{D}-1\right)} exp\left(-\beta s^{\frac{2}{D}}\right), \qquad \text{(Equation 1)}$$

where D is the dimensionality of the space over which the Gaussian process is defined, s is the field size (length in one dimension, area in two dimensions, and volume in three dimensions; 1D, 2D, and 3D hereafter), and the single parameter β can be expressed in terms of σ and θ (see Methods S3, Equation S18). In the 1D case $(D = 1)$, this expression takes the form of a Rayleigh distribution (Figure 1C). Similarly, field gap statistics assume a universal form in the high-threshold limit: threshold crossings are spatially uncorrelated and, consequently, gap sizes follow an exponential distribution with a mean that can be expressed in terms of σ and θ (see STAR Methods). Thus, according to the model, the empirical distributions of CA1 field sizes and gaps between fields should be jointly explained using the two parameters, σ and θ.

We began by examining place fields recently recorded in bats flying in a 200-m-long tunnel, where the highly stereotyped trajectory of the animals along the 1D track allowed for the gathering of comprehensive statistics on the spatial arrangement of place fields.[21] We determined the two parameters σ and θ by matching the means of the field sizes and gaps to the model in the data from the 200-m-long tunnel. This fitting procedure, based only on the means, was sufficient to capture the full distributions of field sizes and gaps (Figures 1C and 1D). This outcome was particularly striking for the field sizes, whose distribution had a distinctive, highly asymmetric form.

Due to its highly asymmetric structure (Figure 1C), the field-size distribution was previously fitted heuristically to a log-normal distribution, yet this approach lacked a principled rationale. By contrast, in our model, the asymmetry is explained by the statistical dependencies between locations of adjacent threshold crossings of the underlying random Gaussian process[34] (see Methods S4). The empirical field-size distribution was more likely to arise from the Gaussian threshold-crossing model than the best-fit log-normal distribution, even though the model had one fewer degree of freedom (assessed via likelihood and parameter-aware likelihood based model performance metrics: $\Delta LLR < 0$, $\Delta BIC < 0$, $\Delta AIC < 0$; see STAR Methods). Furthermore, skew and kurtosis of the log-field-size distribution were in agreement with the threshold-crossing model, but not with the log-normal distribution, according to which they should be close to zero (Figure 1E; see STAR Methods). Thus, the model captured subtle features of the distribution beyond its first- and second-order moments. As expected in the high-threshold limit, the distribution of gap sizes followed an exponential distribution (Figure 1D).

We subsequently tested the model's ability to explain the statistics of place fields measured in rodents running in long 1D tracks. Field size distributions extracted from experiments in which mice ran in a 40-m-long 1D virtual track[24] were in qualitative agreement with the Rayleigh distribution (Figure 1F). We also analyzed the distribution of field sizes from an experiment in which rats explored a 48-m-long 1D maze.[10] The distribution of field sizes was in quantitative agreement with the distribution predicted by the model, with matching skew and kurtosis (Figure S1).

Place fields in two and three dimensions

Experimental data on place fields in rats navigating large 2D environments[25] and bats in 3D rooms[23] reveal place cells with multiple, heterogeneous place fields. Intriguingly, though, the experiments uncover field-size distributions that are distinct in one,

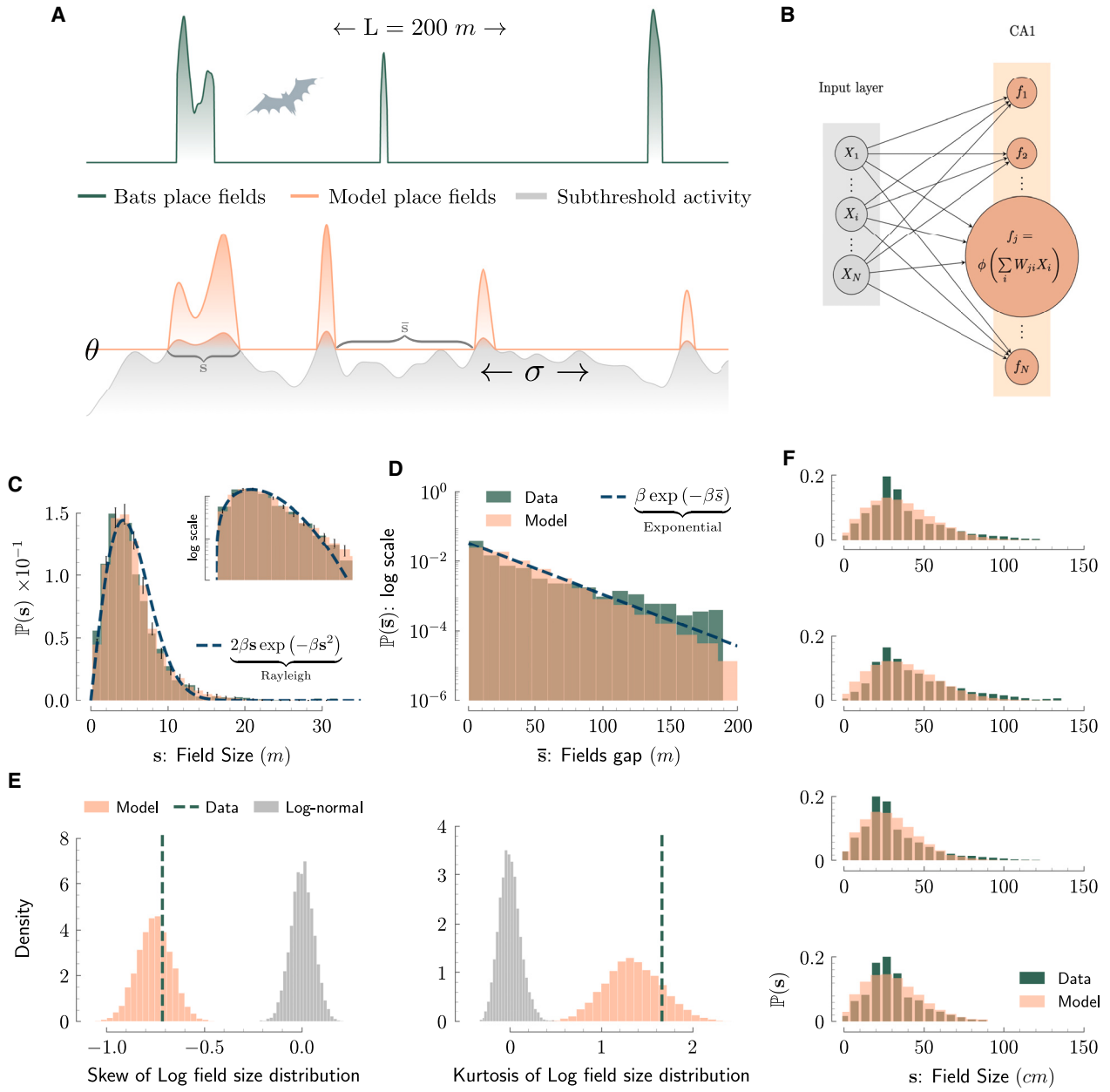


Figure 1. Field arrangements of 1D place fields are explained by the thresholded Gaussian process model

(A) Top: example of place fields measured in bats flying in a 1D 200-m-long tunnel.[21] Bottom: firing rates in the model are generated by thresholding and rectifying a realization of a random Gaussian process (gray). In all figures, green represents experimental data, and orange represents the model.

(B) A simple neural architecture that gives rise to place fields that follow the thresholded Gaussian process model. Presynaptic inputs from many spatially selective neurons are summed with random synaptic weights to create the input to each CA1 place cell, followed by rectification.

(C) Field-size distribution of bats in the 1D tunnel compared against the field-size distribution predicted by the model. Dotted line: approximate analytical prediction of the field-size distribution, valid for high thresholds (Rayleigh distribution). Orange bars: precise predictions of the model, obtained from simulations (error bars: standard deviation across simulations, see STAR Methods). Here and in all subsequent bar plots, the overlap between green bars (experiment) and orange bars (model) is represented in brown. Inset: same data in a semi-logarithmic scale.

(D) Distribution of consecutive field gaps in the experiment compared against the distribution predicted by the model. Dashed line: analytical prediction.

(E) Skew (left) and kurtosis (right) of the log-field-size distribution, obtained from multiple simulations of the model and the best-fit heuristic log-normal distribution (gray). Each simulation is matched to the size of the experimental dataset. For the log-normal distribution, the empirical skew and kurtosis are expected to be close to zero. Dashed line: skew and kurtosis of the experimental distribution.

(F) Rayleigh distribution compared against place-field size distribution in four different mice, running in a 40-m-long virtual track.[24]

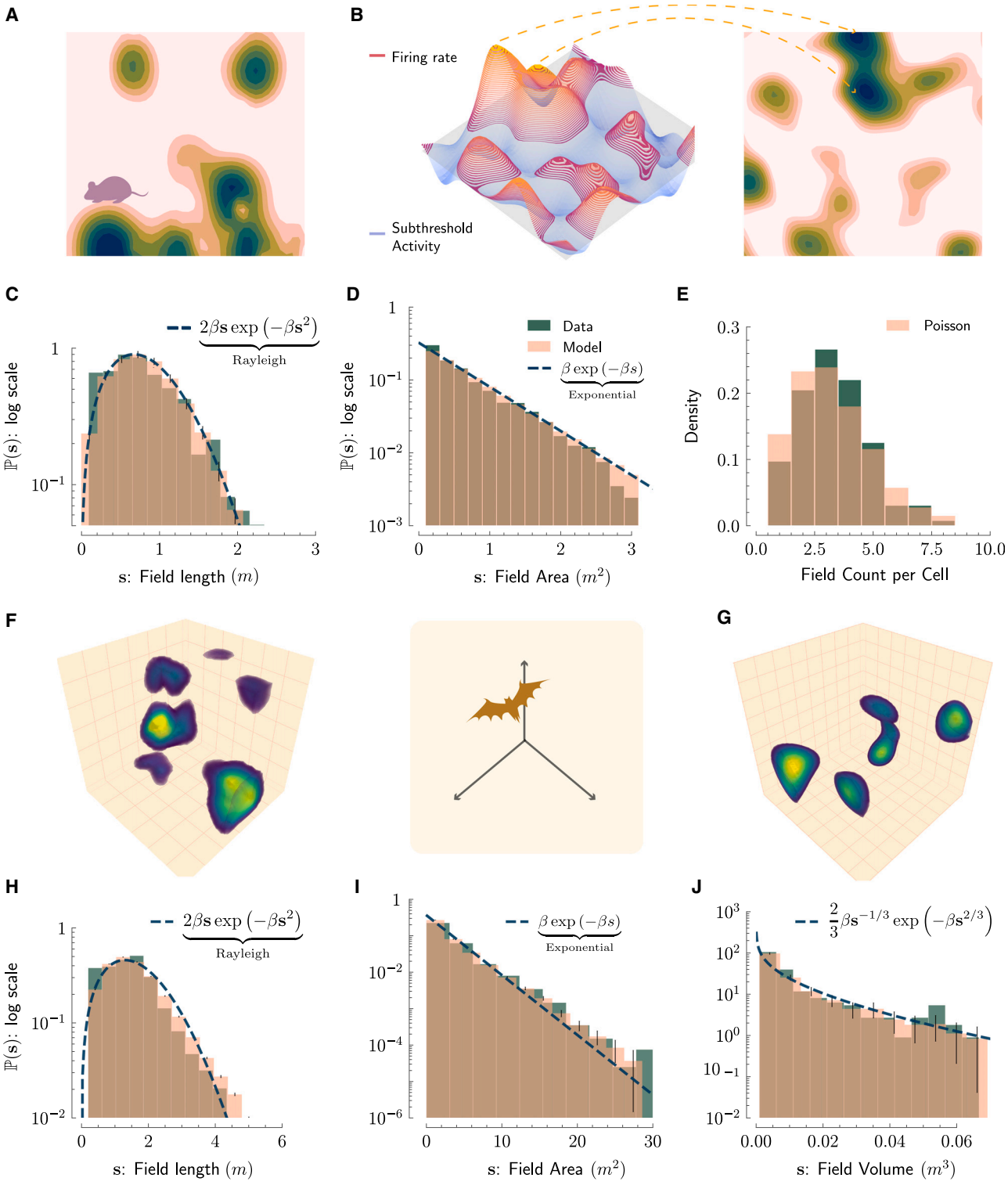


Figure 2. Field arrangements of place fields in 2D and 3D spaces are explained by the thresholded Gaussian process model

(A–E) Model fits to 2D data from rats.

(A) Sample spatial firing pattern of a CA1 cell recorded in a rat navigating in an 18.6 m^2 environment.[25]

(B) Model: random Gaussian process in 2D (left) is thresholded to generate firing fields, yielding the cell's spatial firing pattern (right). For all depictions of distributions, the vertical scale is logarithmic and the color scheme for comparison between the model and the experiment is as in Figures 1C and 1D.

(C) Distribution of field lengths in 1D slices of 2D place fields (20 sample cells from Harland et al.[25]).

(legend continued on next page)

two, and three dimensions. Building on our initial findings in 1D environments, we examined the extension of the model to 2D and 3D environments, where place fields are determined by excursion sets of a Gaussian process over the higher-dimensional space. As in the 1D case, the model generated multiple, heterogeneous fields, which were qualitatively similar to the ones observed experimentally (Figures 2A, 2B, 2F, and 2G).

Moreover, 1D slices through a multidimensional Gaussian process are themselves Gaussian processes. Hence, field sizes in 1D slices through multidimensional firing fields are expected to follow the same field-size distribution as in the 1D case. To test this prediction, we examined 1D slices through 2D place fields measured in rats foraging in a 18.6 m^2 arena (data from 20 cells recorded in Harland et al.[25]) and found that they were well fitted by the Rayleigh distribution (Figure 2C). We further compared the model against a histogram of field areas from all cells (as reported in Harland et al.[25]). The statistics precisely followed an exponential distribution as predicted by the model (Equation 1, $D = 2$; Figure 2D). Finally, field counts were well fitted by a Poisson distribution, as expected in the high-threshold regime in which field locations are independent (Figure 2E).

Next, we analyzed 3D place fields in bats navigating a large room of size 5.8 × 4.6 × 2.7 m^3.[23] The model predicts that 2D slices through the 3D place fields should exhibit the same statistics as those of fields in 2D environments, since these slices are also realizations of a thresholded Gaussian process in two dimensions. The predictions for both 1D and 2D slices were verified by the data (Figures 2H and 2I), along with the distribution of the 3D field volumes (Figure 2J), which followed the prediction of Equation 1 with $D = 3$. In an alternative model, in which field volumes and peak firing rates were exactly matched to the empirical data, without the underlying Gaussian statistics, distributions of 1D and 2D slice sizes did not match the data (Figures S2A and S2B).

In summary, the statistics of field arrangements in 1D, 2D, and 3D were explained in both bats and rodents by the threshold crossings of Gaussian processes. In particular, the model explained the different field-size statistics that were observed across experiments in different dimensionalities (Equation 1; Figures 1C, 2D, and 2J). The generality of these results suggests that common mechanisms, shared across species and dimensionalities, underlie the structure of the hippocampal code for space.

Statistics of field shapes

The statistics of field sizes and gaps only reflect the statistics of field boundaries, which, in the model, correspond to threshold crossings. The model also makes quantitative predictions on the statistics of field shapes, defined as the firing rate's dependence on position within place fields (Figure 3A). These statistics are determined by the properties of the Gaussian process in the threshold-crossing segments, yielding new theoretical predictions that can be tested in the experimental data. First, according to the model, the firing rate within a given place field may exhibit multiple local maxima, unlike in the classical picture of bell-shaped fields. The predicted distribution of the number of local maxima across fields was in close agreement with the empirical distribution of local maxima per field in bats flying in the 1D tunnel (Figure 3B). Second, the model predicts a positive correlation between the size of a field and its peak firing rate, with an approximate power law relation between these two quantities. The experimental data confirmed this prediction, in both bats flying in the 1D tunnel (Figure 3C) and bats flying in the 3D environment (Figure 3D). In both cases, the model not only predicted the qualitative relation between field size and peak firing rate but also accurately predicted the power law coefficient (Figures 3C and 3D). Third, in the model, the slope of the firing rate at the boundaries of 1D place fields follows a Rayleigh distribution.[35] This prediction was verified in the data (Figure 3E). Fourth, for bats flying in the 3D environment, the mean and Gaussian curvatures of isosurfaces formed by the boundary of 3D place fields were distributed in agreement with the predictions of the model (Figures S2D and S2E). All the above predictions of the model were verified using the values of σ and θ that were previously inferred based only on field arrangements, independent of field shapes.

We next examined whether the model captures the topological properties of the place fields by using recent results on the Euler characteristic (EC) of excursion sets of Gaussian processes. The EC of the thresholded Gaussian process is a function of the number of connected components and holes of various dimensions (see STAR Methods) and has a universal dependence on the threshold, which is specific to Gaussian processes.[30] An analytical expression has been obtained for this quantity in terms of θ and σ. (This result has also been used to test for Gaussianity in the spatial structure of the cosmic background microwave radiation, as well as other natural processes.)[36–39]

In the 1D case, the EC is simply equal to the number of connected components. The theory provides an exact expression for the EC curve (Equation 8), whose only dependence on the correlation function is through the correlation length σ. We tested this prediction for place fields by placing a varying threshold on the firing rates measured in bats flying in the 200-m-long tunnel, followed by evaluation of the EC of the rectified fields. The resulting EC curve (Figure 3F) was in excellent agreement with the analytical prediction using the previously obtained value of σ. While an increase in the threshold can be artificially applied to the measured rate maps, it is also possible to reduce the threshold by optogenetically depolarizing during behavior.

(D) Distribution of place-field areas from all cells in the experiment (experimental histogram adapted from Harland et al.[25]).
(E) Distribution of field count per cell across the whole population of recorded cells (experimental histogram adapted from Harland et al.[25]).
(F–J) Model fits to 3D data from bats.
(F) Sample 3D firing pattern recorded in a bat CA1 cell during flight in a 5.8 × 4.6 × 2.7 m^3 room.[23]
(G) Visualization of 3D receptive fields generated by the model.
(H) Distribution of field lengths in 1D slices of bat 3D place fields (analysis performed on all recorded cells in Yartsev and Ulanovsky[23]).
(I) Distribution of field areas in 2D slices of the bat 3D place fields.
(J) Distribution of 3D field volumes in bats. As the dimension increases, the distribution becomes more concentrated around zero and develops a longer tail.
All error bars represent standard deviation across simulations (see STAR Methods).

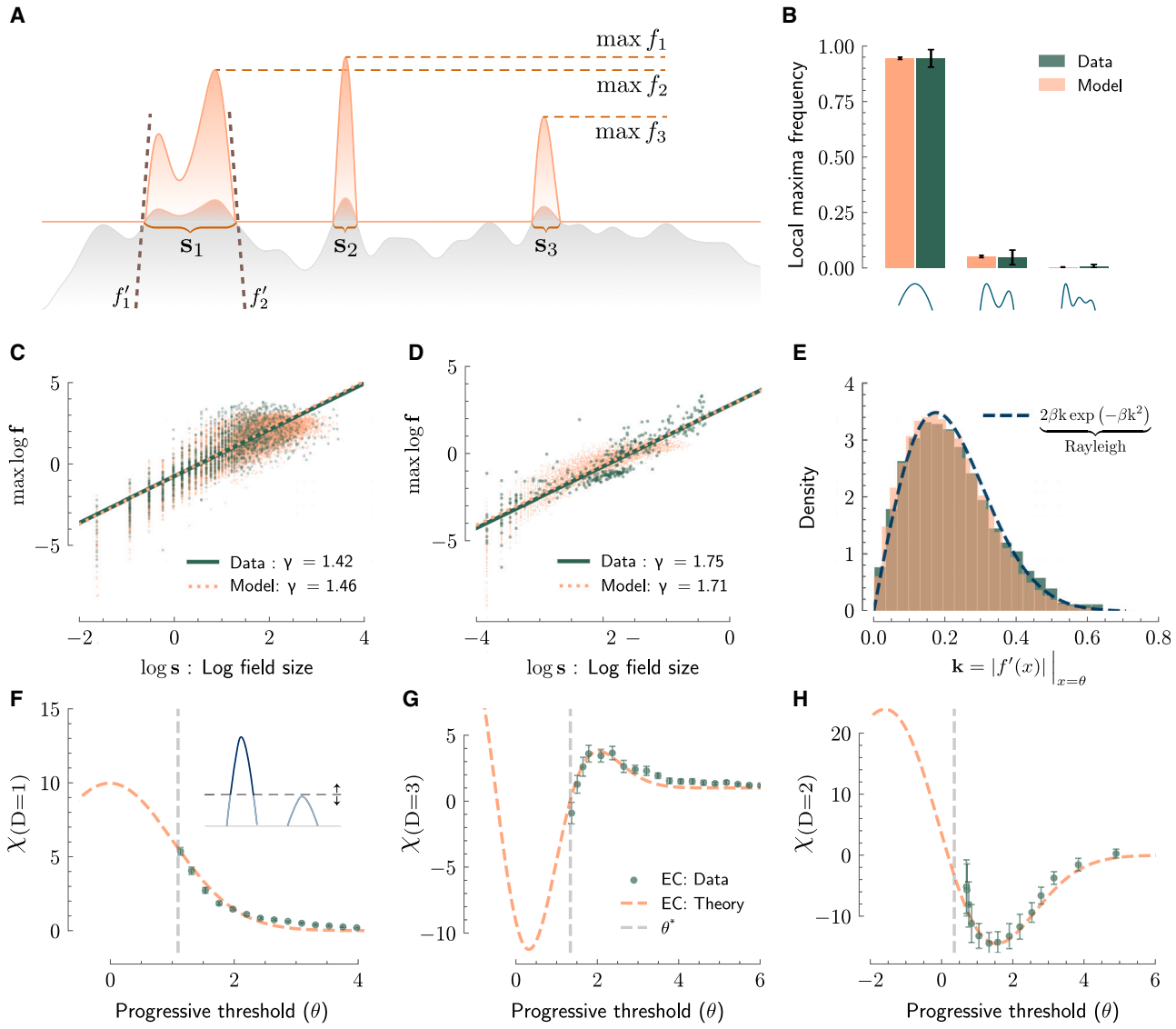


Figure 3. Field-shape-related statistics of place fields are predicted by the thresholded Gaussian process model across dimensions and species

(A) Visualization of various field-shape-related measurement in the case of 1D fields.

(B) Distribution of the number of peaks in the place fields of bats in the 200-m-long tunnel[21] compared with the model. Error bars: standard deviation across simulations (model) and standard deviation across a range of smoothing parameters (data, see STAR Methods).

(C) Joint distribution of log receptive field size (in m) and log maximal firing rate (in Hz) obtained from the model (orange) and from experimentally measured bat place fields in the 200-m-long tunnel (green). Lines: power law relationship between the two quantities in the model and experiment. The coefficient γ is the slope of the best linear fit to the distribution and represents the power law coefficient. The y intercept of the fit is matched between data and model to infer the linear gain factor.

(D) Same as in (A), for place fields recorded in bats flying in the 5.8 × 4.6 × 2.7 m^3 room.[23] Field size is the cubic root of the place-field volume (in m).

(E) Absolute value of field slopes at their boundaries follow a Rayleigh distribution.

(F) EC (χ) as a function of extra thresholding of the place fields of bats in the 200-m-long tunnel (dashed line: theoretical prediction, dots: data). Inset illustrates how adding an extra threshold (horizontal dashed line vs. solid line) can affect the number of fields.

(G) Similar plot as (D) for bats in 3D environment.[23]

(H) Same as (F) and (G) in the 18.6 m^2 2D environment (using 20 sample cells from rats[25]).

Error bars in (F) and (G) represent standard deviation across neurons.

Recent results from such experiments[40] are in qualitative agreement with the threshold-crossing model since the stimulation led to the emergence of unmasked place fields.

In dimensions higher than one, the EC depends not only on the number of connected components but also on the number of holes of various dimensions within the fields. This results in a

A

Dim	Size	σ	θ
1d (m)	6	1	2.4
	130	2.2	1.2
	200	2.4	1.1
2d (m^2)	0.5	0.1	2.2
	2.2	0.2	1.6
	8.2	0.3	1.2
	18.6	0.4	1.2
3d (m^3)	72.0	4.0	1.3

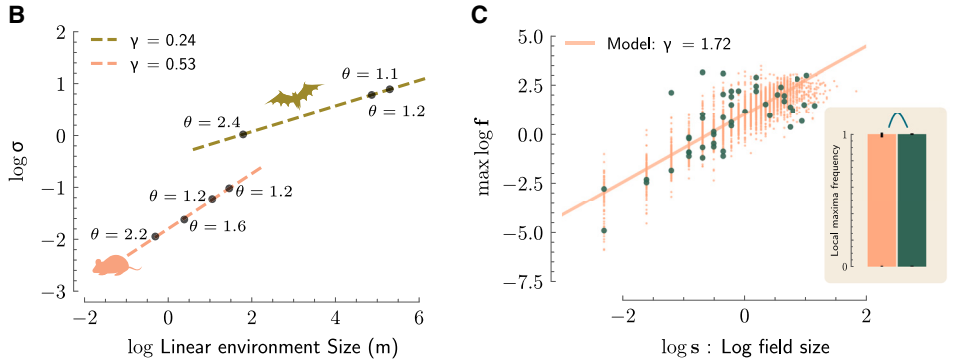


Figure 4. Parameter trends across experiments explain differences in heterogeneity

(A) Summary of parameters for the various experiments in spaces of varying dimension: 1D, bats flying in a tunnel; 2D, rats navigating a large open arena; and 3D, bats flying in a room.

(B) Linear scaling of log σ with the logarithm of the linear environment size in 1D (bats) and 2D (rats).

(C) Joint distribution of log receptive field size, s (in m) and log maximal firing rate, f (in Hz), plotted as in Figure 3C, but in the 6-m-long tunnel. In comparison with the 200-m-long tunnel (Figure 3C), the exponent characterizing the relationship between field size and maximal firing rate is closer to 2, indicating that field shapes in the 6-m-long tunnel are more stereotyped than in the 200-m-long tunnel, as expected due to the higher threshold. Inset: fields with multiple peaks are practically absent, as predicted (compare with Figure 3B).

In (B) and (C), γ is the slope of the best linear fit to the data in logarithmic scales.

non-monotonic dependence of the EC curve on the threshold due to the emergence and disappearance of holes. Here, too, the theory provides an exact analytical expression (Equation 10) for the expected EC curve. We tested this prediction on all measured firing fields in bats flying in 3D.[23] The empirical EC curve was non-monotonic as predicted and precisely followed the analytical expression (Figure 3G) using the previously obtained value of σ. This agreement is specific to the Gaussian threshold-crossing model and was not reproduced by an alternative model with matching joint statistics of field volumes and peak rates (Figure S2C). We also tested the EC curve prediction in 2D on 20 cells from rats recorded in the 2D megaspace[25] and obtained excellent agreement with the analytical prediction (Equation 9; Figure 3H).

Parameter variation across experiments

Our analyses of data from 1D and 2D environments of varying sizes allow us to identify qualitative trends across datasets, as a function of the environment size (Figures 4A and 4B). We observe that the correlation length, σ, increases sublinearly with the size of the environment, and the normalized threshold, θ, decreases with the size of the environment (Figure 4B). Together, these trends induce an increase in the average number of fields with the size of the environment, consistent with the observation of single firing fields per cell in small environments and a sublinear increase in their sizes (Figure S4). It is also interesting to consider the number of fields divided by the environment size, to which we refer as the "average propensity."[24] The sublinear increase of σ and the decrease of θ have opposing influences on this quantity. Consequently, the average propensity varies only weakly with environment size (Figure S3).

While the two parameters jointly affect the field counts and the field sizes, the normalized threshold also influences the heterogeneity in field shapes (Figure 5A). With increasing θ, the fraction of multi-peaked fields decreases and the exponent of the power law relationship between the maximum firing rate and the field size approaches 2, as expected for parabolic fields[30] (Figures 5B and 5C). Thus, the model predicts a greater stereotypy of field shapes with an increasing normalized threshold. As a result, and following the trend seen in the data (Figures 4A and 4B), field shapes are predicted to be more stereotyped in smaller environments than in larger ones (Figures 5A–5C). We tested this prediction on place fields measured in bats flying in a short, 6-m-long tunnel.[21] As predicted, field shapes were more stereotyped than in the 200-m-long tunnel, with statistics that quantitatively agreed with the model (Figures 4C and 5D). In summary, the model accounts for the different characteristics of CA1 spatial selectivity across environmental scales, ranging from early experiments in small environments, where place cells typically had a single, bell-shaped firing field,[8,22] to more recent experiments in large environments, where place cells had multiple firing fields with heterogeneous shapes.[10,21,24,25]

Each one of the experimental datasets is well explained by a single choice of the values of σ and θ, yet some variation in these parameters across cells and anatomical regions is to be expected. For the length scale σ, our choice of a single value is motivated by the fact that recordings were tightly localized anatomically in dorsal CA1.[21,23,25,41] In the data from bats flying in the 1D tunnel, the number of place fields expressed by individual neurons, as well as their sizes, were observed to be correlated in the two flying directions, despite remapping of the fields.[21] We hypothesized that this correlation arises from variability in the threshold, θ. In agreement with this hypothesis, an extended model with a distribution of thresholds accounted for the empirical correlations, while remaining compatible with the distributions of field sizes and gaps obtained from the basic model (Figure S4). A distribution of thresholds may arise from a variation in intrinsic cell properties or from variation in the inhibitory inputs to place cells and is consistent with the recent observation that propensity varies across cells also in rodents.[24]

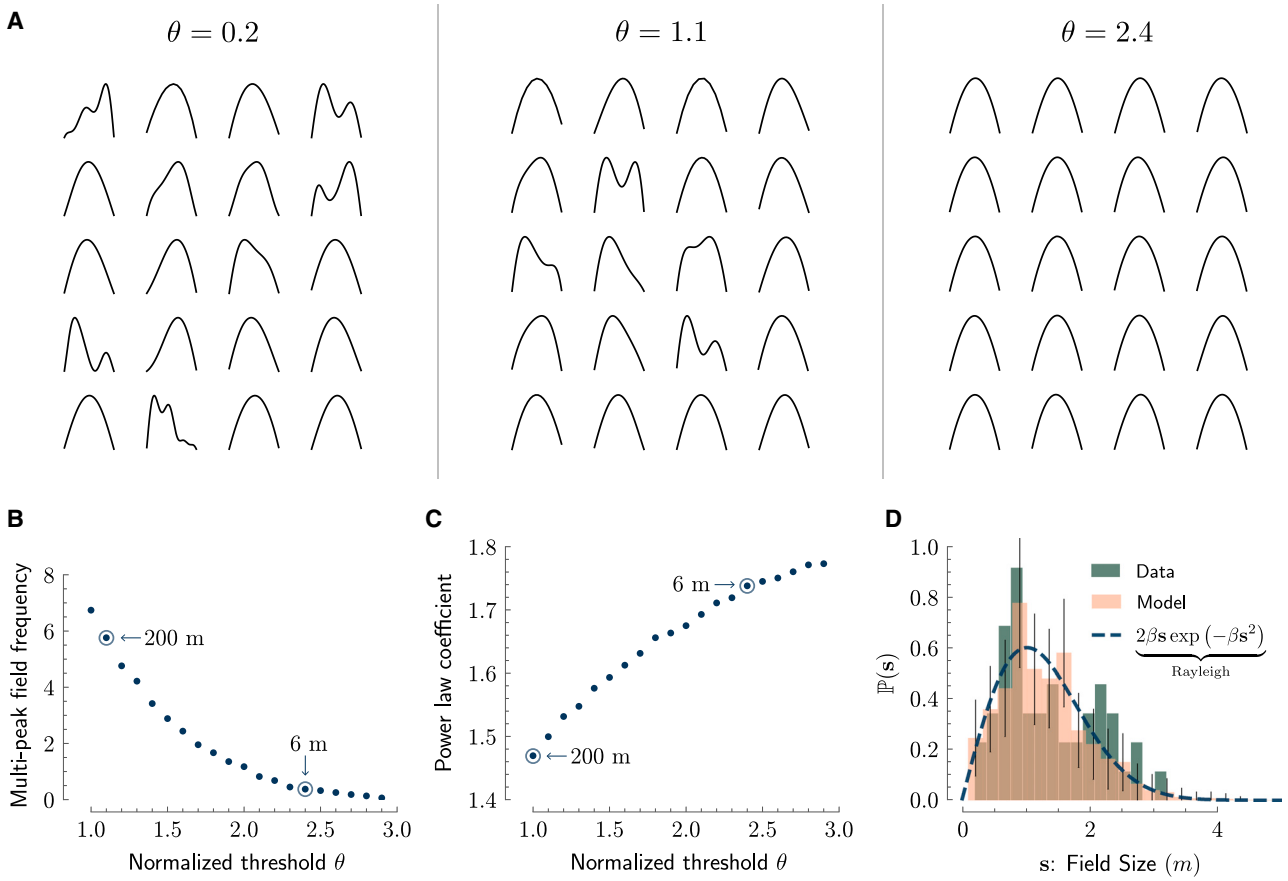


Figure 5. Field shape as a function of threshold

(A) Visualization of randomly selected sample field shapes generated by the model ranging from low- to high-threshold regime (left to right, the intermediate value of θ = 1.1 is the normalized threshold inferred from recordings in the 200-m-long tunnel and the largest value θ = 2.4 is the normalized threshold inferred from recordings in the 6-m-long tunnel).

(B) Predicted percentage of cells with more than one local peak, as a function of the normalized threshold.

(C) Predicted logarithmic slope of field width vs. max. firing rate, as a function of the normalized threshold. It can be shown that in the limit of very high threshold, fields are parabolic and the logarithmic slope approaches 2.[30]

(D) Field widths in bats flying in a 6-m-long tunnel follow a Rayleigh distribution despite more stereotypy in field shapes as predicted (40 recorded cells; peak count in the figure is 8; the distributions have no significant difference; Kolmogorov-Smirnov [K-S] test: p value = 0.6, K-S stat = 0.1). Error bar represents standard deviation across simulations (see STAR Methods).

DISCUSSION

The classical view of spatial coding in the hippocampus has been challenged by the recent discoveries of heterogeneous and distributed CA1 response patterns in large environments. It was unknown whether a unified mathematical framework can encompass the diverse statistics of these irregular responses. Here, we identified such a framework, in which place fields are obtained from the threshold crossings of a spatially fluctuating random process. The underlying randomness of the Gaussian process enabled the model to explain in quantitative detail the variability in the size of place fields and their spatial arrangement, as well as a range of geometric and topological properties associated with the heterogeneous field shapes. Indeed, when constrained only on the mean field size and count, the model precisely captured the statistics of shape variability, which, in principle, could have had an independent origin. This surprising result suggests that the different degrees of heterogeneity, observed across experiments, arise as a byproduct of mechanisms that regulate the field frequency and size. With only two parameters, the model accounts for the seemingly diverse statistics observed in rodents and bats and in environments of varying sizes and dimensionalities. Therefore, the model allows for direct comparison between the statistics observed in the different experiments and suggests that they arise from common underlying principles. Specifically, our analysis of the experimental data indicates that there are systematic differences in the statistics of place fields in environments of different size: field sizes increase with environment size, propensity decreases, and field shapes become more heterogeneous. Consequently, place fields in small environments differ from those observed in a similarly sized portion of a larger environment.

A Gaussian process is the random process that maximizes entropy under constraints on the mean and the spatial covariance function. Hence, the model invokes minimal assumptions other

than a specification of a place cell's underlying input correlation structure.[31,42] At the same time, random Gaussian processes can generate a highly efficient coding scheme, with capacity that increases exponentially with the number of neurons,[29] and, therefore, the structure of tuning curves in CA1 might be shaped to optimize coding efficiency. Another possibility is that the structure of neural responses in CA1 is beneficial for the maintenance and retrieval of associative memory,[27] since CA1 is part of a larger system involved in this function.[43] It will be interesting to examine theoretically whether it is possible to obtain a coherent understanding of the trends seen in the parameters across the different geometries (Figure 4B) in light of a theory of efficient coding[29] or of optimal storage capacity.

The success of the parsimonious model introduced here invites an exploration into its possible mechanistic origin. The apparent stereotypy of place fields in small environments has motivated a view of hippocampal spatial selectivity as arising from highly organized synaptic connectivity.[44–46] However, the irregular CA1 firing patterns in large environments are inconsistent with this view. Our results indicate that CA1 place fields, in small and large environments alike, are compatible with predominantly random synaptic connections into CA1. This is because random projections from a sufficiently large number of spatially selective cells necessarily produce an input to each CA1 cell that varies spatially as a realization of a random Gaussian process, due to the multivariate central limit theorem (see Methods S2). Gaussian statistics arise robustly from random synaptic projections without the necessity to invoke highly specific assumptions on the structure of spatial responses in the input layer. These can range from regular tuning curves that tile the environment to heterogeneous and irregular tuning curves with a distance-dependent covariance function (see Methods S2). Hence, the broad success of the model across datasets and statistics suggests that randomness, rather than specific design, governs the synaptic organization of inputs to CA1.

Randomness in the connectivity might be associated with a fairly fixed anatomical structure, potentially influenced by the developmental trajectory of neurons and their embedding in the network.[47] Yet, the strength of synaptic projections between CA3 and CA1 is known to be plastic. Specifically, recent experiments indicate that a dominant mode of plasticity in CA1 involves randomly occurring modifications of synaptic projections from CA3, triggered by inputs from the entorhinal cortex.[48–50] The randomness of the synaptic weights may be related to the cumulative effect of many such non-Hebbian plasticity events. Since all the measurements analyzed in this work were collected in highly familiar and uniform environments, it is possible that the stationarity of the underlying Gaussian processes is specific to such conditions. We speculate that more intricate features of the place-cell code arise under richer behavioral conditions via activity-dependent plasticity mechanisms acting on top of a scaffold of random synaptic weights.

RESOURCE AVAILABILITY

Lead contact

Requests for further information and resources should be directed to and will be fulfilled by the lead contact, Yoram Burak (yoram.burak@elsc.huji.ac.il).

Materials availability

This study did not generate new unique reagents.

Data and code availability

- This paper analyzes existing, publicly available data, accessible at https://doi.org/10.5281/zenodo.46467270 and http://amygdala.psychdept.arizona.edu/lab.html.
- All original code is publicly available at https://doi.org/10.5281/zenodo.14650880.
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

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AUTHOR CONTRIBUTIONS

N.M. and Y.B. conceptualized the research and developed the methodologies, with major inputs from R.A.d.S. All authors interpreted the results. N.M. developed the code for simulation and data analysis, performed simulations, and analyzed the data. N.M. and Y.B. wrote the first draft of the manuscript with major inputs from R.A.d.S. All authors participated in writing and editing of the manuscript. Y.B. supervised the project and acquired funding.

DECLARATION OF INTERESTS

The authors declare no competing interests.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Deposited data		
Datasets	Eliav et al.[21]	https://doi.org/10.5281/zenodo.4646727
Datasets	Harland et al.[25]	http://amygdala.psychdept.arizona.edu/lab.html
Software and algorithms		
Python 3.11	Python Software Foundation	https://www.python.org/
Code for this paper	https://doi.org/10.5281/zenodo.14650880	https://github.com/5183nischal/gp-model

METHOD DETAILS

Mathematical notation

Throughout the main text, STAR Methods section, and supplemental information, we use the mathematical notation defined below:

Symbol	Description
L	Length of one-dimensional (1D) environment
D	Dimension of space
L^D	Volume of environment for $D > 1$
x	Spatial coordinate
$\mathcal{GP}$	Gaussian process
$h(x)$	Gaussian process input to CA1 cell
$r(x - x')$	Covariance function of $h(x)$
$r(0)$	Variance of $h(x)$
$-r''(0)$	Variance of $h'(x)$, the derivative process
θ	Normalized threshold
σ	Correlation length
$f(x)$	Firing rate of CA1 cells in the model
$\phi(\cdot)$	Gaussian cumulative density function
N	Field count per cell
s	Field size
$\bar{s}$	Field gap
χ	Euler characteristic

Model details

Place fields are modeled as rectified realizations of a stationary Gaussian process over D dimensional space ($x \in \mathbb{R}^D$) with 0 mean and stationary covariance function $r(x - x')$,

$$h(x) \sim \mathcal{GP}[0, r(x - x')]. \quad \text{(Equation 2)}$$

A sample $h(x)$ from the Gaussian process can be thought of as input to a CA1 place cell. The input is subsequently thresholded at a value $\theta\sqrt{r(0)}$ to generate the cell's firing rate, which is proportional to $f(x)$, defined as

$$f(x) = \max\left(h(x) - \theta\sqrt{r(0)}, 0\right), \quad \text{(Equation 3)}$$

where θ is the normalized threshold, equal to the ratio between the threshold and the standard deviation of the process. The correlation length of the Gaussian process is defined as[30]:

$$\sigma = \sqrt{-\frac{r(0)}{r''(0)}}. \quad \text{(Equation 4)}$$

Thus, σ is equal to the ratio between the standard deviation of the process $\sqrt{r(0)}$ and the standard deviation of its derivative $\sqrt{-r''(0)}$.[30] For simulations of the Gaussian process, we used a transitionally invariant Gaussian correlation function.

QUANTIFICATION AND STATISTICAL ANALYSIS

Data processing

Place field firing rate maps (post place field identification) in bats in 1D[21] and 3D[23] environments were obtained and rectified at $\approx$ 0.5 Hz for denoising. In the 3D case, firing rates also went through a median filtering step using *scipy.ndimage* Python library with minimal (nearest neighbor) kernel of size 2 (place fields resolution = 10 cm on each axis) to eliminate salt-and-pepper noise while preserving field boundaries.

Parameter inference in 1D environments

The two parameters σ and θ were inferred by matching the means of the field size and the field gap to those in the model via simulation in the dataset from bats in the 200 m[21] (Figures 1C, 1D, 4A, and 4B). In the dataset from bats in the 6-m-long tunnel (Figure 5D), most cells had a single field and parameters were inferred by matching the mean field size and the mean fraction of the environment in which place cells were active. For the dataset from bats in the 130-m-long tunnel and dataset from rats in the 48-m-long maze, mean field size and mean field count were matched to infer the parameters[10,21] (Figures S1, 4A, and 4B). For rats in a 40-m-long virtual track[24] (Figure 1F), where only a histogram of field sizes was available, we directly fitted the parameter β of the approximate Rayleigh distribution (Equation 1). Parameters matched through simulation yielded excellent agreement with the analytical formulas in the 1D case (see Methods S3, Equations S21 and S22).

Parameter inference in 2D environments

For rats in the 2D environments[25] (Figures 1C–1E, 4A, and 4B), parameters were inferred by fitting the model to the mean field count and mean field size reported in Harland et al.[25] When matching the mean count, we took into account the systematic exclusion of cells with zero fields from experimental observations by fitting the model to the distribution of field counts, conditioned on the existence of at least one field. For the Poisson distribution (valid in the high threshold regime), the mean $\langle N\rangle_{\text{experiment}}$ of the conditioned distribution, which is matched to the experimental data, is then

$$\langle N\rangle_{\text{experiment}} = \frac{\langle N\rangle}{1 - \exp(-\langle N\rangle)} \quad \text{(Equation 5)}$$

where $\langle N\rangle$ is the unconditioned mean. Analysis of the Euler Characteristic in the large 2D environment[25] (Figure 2H) was performed on a sample of firing rate maps from 20 cells shared with us by the authors of the manuscript. All other 2D results are based on statistics from the full dataset, as reported in Harland et al.[25]

Parameter inference in 3D environment

For bats in a 3D environment[23] (Figures 2H–2J and 4A) the parameters σ and θ were inferred by matching the mean field volume and the mean fraction of the volume in which place cells were active, based on the full rate maps.

Fields in 1D

Fields were defined as connected components of non-zero activity, and gaps as connected components of zero activity (Figures 1C, 1D, 2B, and 2E). When generating histograms based on the model (Figures 1C, 2C, 2H–2J, and 4C), errors bars were obtained by performing 10^4 simulations of random fields using the model, each matching the number of cells to the data. Field size distributions from model simulations in 1D were compared against the best fit log-normal distribution by first generating a kernel density estimate of the field size distribution from the simulations (*scipy.stats.gaussian_kde*). The model has one less degree of freedom than the two-parameter log-normal distribution, since constraining the model parameters to also fit the gap size distribution reduces the parameter space to one dimension. Nevertheless, the log likelihood test indicated that the model fits the data better than the log-normal distribution ($\Delta\text{LLR} = -50.3 < 0$). Differences in model complexity aware log likelihoods were measured using the Bayesian Information (BIC) and Akaike Information (AIC) criteria, yielding $\Delta\text{BIC} = -109.1 < 0$ and $\Delta\text{AIC} = -104.5 < 0$, which implies that the data fit the model better than log-normal.

Kurtosis and skew

Kurtosis, quantifying the heaviness of distribution tails, was defined as

$$\text{Kurtosis} = \frac{m_4}{m_2^2} - 3, \quad \text{(Equation 6)}$$

where m_4 and m_2 are the fourth and second moments of the distribution of log-field sizes, respectively. Skew, quantifying the asymmetry of the distribution was defined as

$$\text{Skew} = \frac{m_3}{m_2^{3/2}} \quad \text{(Equation 7)}$$

where m_3 and m_2 are the third and second moments of the log-field-size distribution, respectively. Both measures vanish for a log-normal distribution, since in that case the logarithrm of field sizes is normally distributed. The effect of rectification as part of data processing is explored in Figure S5A.

Fields in 3D

Field statistics in 3D were measured from the firing rate data from Yartsev and Ulanovsky.[23] To measure field volumes, connected components in the 3D volumetric fields were identified using the Python function *scipy.ndimage.label*, which identifies and labels connected regions in N-dimensional arrays. Fields with volume larger than 3 standard deviations from the mean were removed as outliers, as they are likely to result from merging of two or more fields. These large outliers were also removed when analysing the slices of the 3D fields (Figures 2D–2F).

Number of local peaks within a field

Maxima in each field from 1D data were identified using the Python function *scipy.signal.find_peaks*, which finds local peaks in a 1D array. Incomplete fields at boundaries were removed. To filter out spurious maximas in the data arising from noisy assessment of the firing rate, we first applied a Gaussian smoothing with a standard deviation ranging from 0.5 to 1 m in the case of bats flying in a 200-m-long tunnel, and Gaussian smoothing with a standard deviation ranging from 0.3 to 0.6 m in the case of bats flying in a 6-m-long tunnel. Local peak count distribution was calculated for the smoothed fields in each case and mean and standard deviation are reported. The error bars in the model represent the standard deviation across independent simulations (Figures 3B and 4C).

Maximum firing rate vs. field width

Maximum firing rates in Figures 3C and 3D were defined as the maximum rate observed in each field within its connected region. In the 1D data (Figure 3C), fields were rectified by a small threshold of 1 Hz to eliminate spurious measurements arising from regions in which the firing rate measurement is unreliable (note that the Gaussian threshold crossing model is applicable for the rectified data). The power law coefficient extracted from the data was insensitive to the choice of the threshold above this value (Figure S5B). The y intercept of the linear fit on the log-log plot was matched to eliminate the effect of the linear gain factor.

Boundary slopes

Field boundary slopes were extracted by fitting a line between the two consecutive samples of the firing rate (separated by 0.1 m) at the boundary of each rectified field, and measuring its slope (Figure 3E).

Euler characteristics

The Euler characteristic (EC) is a topological invariant that can be expressed as the alternating sum of Betti numbers. The 0th order Betti number is equal to the number of connected components in a topological space, and the n th order Betti number is equal to the number of n -dimensional holes. The EC χ of a Gaussian process in D dimensions has a universal form, insensitive to the structure of the underlying covariance function (under mild regularity requirements[30]):

$$D = 1 \Rightarrow \langle \chi \rangle = \left[\frac{(L/\sigma)}{2\pi}\right] e^{\frac{-\theta^2}{2}} + (1 - \phi(\theta)), \quad \text{(Equation 8)}$$

$$D = 2 \Rightarrow \langle \chi \rangle = \left[\frac{(L/\sigma)^2}{(2\pi)^{3/2}}\theta + \frac{2(L/\sigma)}{2\pi}\right] e^{\frac{-\theta^2}{2}} + (1 - \phi(\theta)), \quad \text{(Equation 9)}$$

$$D = 3 \Rightarrow \langle \chi \rangle = \left[\frac{(L/\sigma)^3}{(2\pi)^2}(\theta^2 - 1) + \frac{3(L/\sigma)^2}{(2\pi)^{3/2}}\theta + \frac{3(L/\sigma)}{2\pi}\right] e^{\frac{-\theta^2}{2}} + (1 - \phi(\theta)), \quad \text{(Equation 10)}$$

where ϕ is the Gaussian CDF function. Empirical Euler characteristics (EC) were measured using the *Gudhi* python library by taking the alternating sum of Betti numbers. The mean maximum firing rate was normalized in the data to match that of the model to determine the linear gain value before calculating the EC.